Answer Guide for the Chapter Challenges

Chaos: An Introduction to Dynamical Systems
(Alligood, Sauer and Yorke)
Challenges 3, 4, 5, 11, 12 and 13

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Equation and theorem numbers refer to the textbook. Each challenge is solved in the order in which the book poses its steps, and every step ends where the next one picks up. Challenges 1 and 2 are not included, and neither are Challenges 6 to 10.

3 Challenge 3: Sharkovskii's Theorem

Setting. Throughout, f is a continuous map of an interval and $P = \{x_1 < x_2 < \cdots < x_p\}$ is a periodic orbit of least period $p \geq 3$, so f permutes P in a single cycle. The *partition intervals* are $I_i = [x_i, x_{i+1}]$ for $1 \leq i \leq p-1$. They overlap only at endpoints. Write $I \to J$ (an arrow of the transition graph) when $f(I) \supseteq J$.

The image of an interval is an interval. If $I = [x, y]$ is a partition interval, then $f(I)$ contains the segment between $f(x)$ and $f(y)$, which are distinct points of P. So $I \to J$ for every partition interval J lying between $f(x)$ and $f(y)$.

Strategy. Steps 1 to 4 extract structure from a single period-p orbit. Steps 5 to 8 turn that structure into loops in a transition graph, and loops give periodic orbits. Steps 9 to 11 handle the powers of two and assemble the theorem. Two lemmas on loops are used throughout.

Lemma A (loops). If $I_{j_0} \to I_{j_1} \to \cdots \to I_{j_{n-1}} \to I_{j_0}$ is a loop in the transition graph, then there is a point $x \in I_{j_0}$ with $f^t(x) \in I_{j_t}$ for $0 \le t < n$ and $f^n(x) = x$.

Proof. First, if $f(I) \supseteq J$ for closed intervals I, J, then some closed subinterval $I' \subseteq I$ has $f(I') = J$. To see this write $J = [c, d]$ and pick $s, t \in I$ with $f(s) = c$ and $f(t) = d$, say $s < t$ (the other case is symmetric). Let t' be the smallest point of $[s, t]$ with $f(t') = d$ and s' the largest point of $[s, t']$ with $f(s') = c$. By the intermediate value theorem f cannot leave $[c, d]$ on $[s', t']$, so $f([s', t']) = J$.

Now put $K_n = I_{j_0}$ and work backward, choosing closed intervals $K_t \subseteq I_{j_t}$ with $f(K_t) = K_{t+1}$ for $t = n-1, \ldots, 0$. This is possible because $f(I_{j_t}) \supseteq I_{j_{t+1}} \supseteq K_{t+1}$. Then $f^n(K_0) = I_{j_0} \supseteq K_0$, and the Fixed-Point Theorem (Theorem 3.17) applied to f^n gives $x \in K_0$ with $f^n(x) = x$. Also $f^t(x) \in K_t \subseteq I_{j_t}$. □

Lemma B (least period). Let x be as in Lemma A and suppose the cyclic sequence $(I_{j_0}, \ldots, I_{j_{n-1}})$ is not a repetition, meaning it has no period $m < n$ when indices are read modulo n. Then either x has least period exactly n, or $x \in P$ (so $p \mid n$). In particular x has least period n whenever $p \nmid n$.

Proof. Let m be the least period of x and suppose $m < n$, so $m \mid n$. For each t the point $f^t(x) = f^{t+m}(x)$ lies in both I_{j_t} and $I_{j_{t+m}}$. Distinct partition intervals meet only at points of P. Hence either $I_{j_t} = I_{j_{t+m}}$ for all t, which makes the sequence m-periodic and is excluded, or some $f^t(x)$ lies in P. In the second case $x \in P$ because P is invariant and x is periodic. □

Step 1: The chain $A_1 \subseteq f(A_1) \subseteq f^2(A_1) \subseteq \cdots$

Let $A_1 = [u, w] = [x_i, x_{i+1}]$ be the rightmost partition interval whose left endpoint maps to the right of itself. Such an interval exists because $f(x_1) > x_1$. Since $f(u) \in P$ and $f(u) > u$, we get $f(u) \ge w$. If $i + 1 < p$, the interval starting at w is not of this type by maximality, so $f(w) \le w$. If $i + 1 = p$ then $f(w) < w$ trivially. Since P has no fixed points, $f(w) < w$, hence $f(w) \le u$. Together,

$$f(u) \ge w, \qquad f(w) \le u, \qquad \text{so} \qquad A_1 \subseteq f(A_1). \tag{1}$$

If $A \subseteq B$ then $f(A) \subseteq f(B)$. Applying f to $A_1 \subseteq f(A_1)$ gives $f(A_1) \subseteq f^2(A_1)$, and inductively $A_1 \subseteq f(A_1) \subseteq f^2(A_1) \subseteq \cdots$. Each term is an interval, being a continuous image of an interval.

Step 2: The chain picks up orbit points one at a time

(a) The statement in the text. Let $P_j = P \cap f^j(A_1)$. By Step 1, $P_j \subseteq P_{j+1}$. Also $f(P_j) \subseteq f^{j+1}(A_1) \cap P = P_{j+1}$. Suppose $P_{j+1} = P_j$. Then $f(P_j) \subseteq P_j$, and since f is injective on the finite set P, $f(P_j) = P_j$. A nonempty f-invariant subset of P is all of P, because P is a single cycle. So the number of orbit points in $f^j(A_1)$ increases strictly until it equals p. It starts at $|P_0| = 2$, hence reaches p for some $j \le p - 2$. At that stage $f^j(A_1)$ is an interval containing x_1 and x_p, so it contains $[x_1, x_p]$.

(b) The version needed for loops. The sets $f^j(A_1)$ may extend beyond what the partition intervals record, so for loops we count intervals reached by *paths*. Let $\mathcal{R}_j$ be the set of partition intervals J for which the transition graph has a path of length j from A_1 to J. Then $f^j(A_1) \supseteq \bigcup \mathcal{R}_j$, by the covering rule.

- $\mathcal{R}_0 = \{A_1\}$ and $\mathcal{R}_j \subseteq \mathcal{R}_{j+1}$, because (1) gives a loop $A_1 \to A_1$ that can be prepended to any path.
- $\mathcal{R}_j$ is a block of consecutive intervals. By induction, if adjacent intervals $I, I' \in \mathcal{R}_j$ share the endpoint z, then the intervals covered by $f(I)$ and by $f(I')$ both include intervals touching the point $f(z) \in P$, so their union is again a block. The union over a block of adjacent intervals is a block.
- Let Q_j be the set of endpoints of intervals in $\mathcal{R}_j$. If $I = [x, y] \in \mathcal{R}_j$, then the intervals between $f(x)$ and $f(y)$ lie in $\mathcal{R}_{j+1}$, so $f(Q_j) \subseteq Q_{j+1}$. Also $Q_j \subseteq Q_{j+1}$. If $\mathcal{R}_{j+1} = \mathcal{R}_j$ then $f(Q_j) = Q_j$ as in (a), so $Q_j = P$ and $\mathcal{R}_j$ is the full set of $p-1$ intervals.

So $|\mathcal{R}_j|$ increases strictly until $\mathcal{R}_j$ contains every partition interval. Let k be the first j with this property. Since $|\mathcal{R}_0| = 1$, we get $k \le p-2$, and $f^k(A_1) \supseteq [x_1, x_p]$. This k is the one used from now on.

Step 3: Another interval covers A_1, or p is even and there is a period-two orbit

Let $\mathcal{L} = \{x_1, \ldots, x_i\}$ be the orbit points on the left side of A_1 (including $u = x_i$) and $\mathcal{U} = \{x_{i+1}, \ldots, x_p\}$ those on the right (including w). By (1), $f(u) \in \mathcal{U}$ and $f(w) \in \mathcal{L}$.

If some partition interval $I_j = [x_j, x_{j+1}]$ has $f(x_j) \in \mathcal{L}$ and $f(x_{j+1}) \in \mathcal{U}$ (or the reverse), then $f(I_j) \supseteq [x_i, x_{i+1}] = A_1$. Suppose no interval other than A_1 covers A_1. Then for every $j \ne i$ the images $f(x_j)$ and $f(x_{j+1})$ lie on the same side. Chaining over $j = 1, \ldots, i-1$ shows that all of $f(\mathcal{L})$ lies on the side of $f(u)$, that is in $\mathcal{U}$. Chaining over $j = i+1, \ldots, p-1$ shows $f(\mathcal{U}) \subseteq \mathcal{L}$. Because f is a bijection of P, $|\mathcal{L}| \le |\mathcal{U}| \le |\mathcal{L}|$, so $|\mathcal{L}| = |\mathcal{U}| = p/2$ and p is even, with $f(\mathcal{L}) = \mathcal{U}$ and $f(\mathcal{U}) = \mathcal{L}$.

In that case let $J_L = [x_1, x_i]$ and $J_R = [x_{i+1}, x_p]$. Since $f(\mathcal{L}) = \mathcal{U}$, the interval $f(J_L)$ contains all of $\mathcal{U}$ and hence J_R. Likewise $f(J_R) \supseteq J_L$. So $J_L \to J_R \to J_L$ is a loop, and Lemma A gives $x \in J_L$ with $f(x) \in J_R$ and $f^2(x) = x$. The two intervals are disjoint, so $f(x) \ne x$, and x has least period 2.

Step 4: Either a period-$(p-2)$ orbit, or a period-two orbit, or $k = p-2$

Suppose $k \le p-3$ and alternative (2) of Step 3 does not occur. By Step 3 there is a partition interval $B \ne A_1$ with $B \to A_1$. Since $\mathcal{R}_k$ contains every interval, B is reachable from A_1. Take a shortest path $A_1 \to J_1 \to \cdots \to J_{d-1} \to B$ of length $d \le k \le p-3$. A shortest path never returns to A_1, so $J_1, \ldots, J_{d-1}, B$ are all different from A_1. Let $r = p-2-d \ge 1$ and consider the loop

$$\underbrace{A_1 \to A_1 \to \cdots \to A_1}_{r \text{ copies}} \to J_1 \to \cdots \to J_{d-1} \to B \to A_1,$$

whose length is $r + d = p - 2$. The arrows $A_1 \to A_1$ and $B \to A_1$ exist, and the others come from the path. The cyclic sequence has exactly one maximal run of A_1's, so it is not a repetition. Lemma A gives a point x with $f^{p-2}(x) = x$. If $x \in P$ its least period would be p, which cannot divide $p-2$. So by Lemma B, x has least period $p-2$, and it lies in $[x_1, x_p]$.

Hence either (1) f has a period-$(p-2)$ orbit in $[x_1, x_p]$, or (2) p is even and f has a period-two orbit, or (3) $k = p-2$. In case (3), $\mathcal{R}_j$ gains exactly one interval at each step, as used below.

Step 5: The picture for $p = 9$

Assume p is the smallest odd period larger than 1. For the example $p = 9$ this means f has no orbit of period 3, 5 or 7. Alternatives (1) and (2) of Step 4 are excluded, so $k = p - 2$. Since $|\mathcal{R}_0| = 1$, $|\mathcal{R}_{p-2}| = p - 1$ and every step adds at least one interval, every step adds exactly one. Write $\mathcal{R}_j = \{A_1, \ldots, A_{j+1}\}$ where A_{j+1} is the new interval at step j. It sits next to the block $B_{j-1} = A_1 \cup \cdots \cup A_j$ on its left or right. Let y_j be the outer endpoint of A_{j+1}. Then $Q_j = \{u, w, y_1, \ldots, y_j\}$. Put $y_{-1} = w$ and $y_0 = u$.

(a) Arrows. For $1 \le i \le p-2$, $A_i \in \mathcal{R}_{i-1}$, so A_i covers only intervals in $\mathcal{R}_i = \{A_1, \ldots, A_{i+1}\}$. The interval $A_{i+1} \in \mathcal{R}_i \setminus \mathcal{R}_{i-1}$ must be covered by some member of $\mathcal{R}_{i-1}$, and members with index below i lie in $\mathcal{R}_{i-2}$ and cover only $\mathcal{R}_{i-1}$. So $A_i \to A_{i+1}$ for $1 \le i \le p-2$.

(b) The cycle. For $1 \le j \le p-2$ the point y_j lies in $f(Q_{j-1})$. Otherwise $f(Q_{j-1}) \subseteq Q_j \setminus \{y_j\} = Q_{j-1}$, so Q_{j-1} would be an invariant proper subset of the cycle. For $j \ge 2$ it cannot be the image of a point of Q_{j-2}, since $f(Q_{j-2}) \subseteq Q_{j-1} \not\ni y_j$. Hence $f(y_{j-1}) = y_j$ for $2 \le j \le p-2$, and $y_1 \in \{f(u), f(w)\}$. The points u and w are the only ones not yet placed. If $f(u) = y_1$, then $f(w) \ne w$ forces $f(w) = u$ and $f(y_{p-2}) = w$. By (1), $f(u) \ge w$ so $y_1 > w$, and A_2 lies to the right of A_1. If instead $f(w) = y_1$ we get the mirror image of everything that follows. So we may assume

$$w \to u \to y_1 \to y_2 \to \cdots \to y_{p-2} \to w,$$

that is $f(y_{-1}) = y_0$, $f(y_i) = y_{i+1}$, $f(y_{p-2}) = y_{-1}$, with A_2 on the right.

(c) The new intervals alternate sides. Let $s_j \in \{L, R\}$ be the side of B_{j-1} on which A_{j+1} lies, with $s_0 = L$ and $s_1 = R$. Suppose the sides alternate up to s_{j-1} and then $s_j = s_{j-1}$ for some $j \ge 2$. By induction B_{j-1} has the endpoint y_{j-1} on side s_{j-1} and y_{j-2} on the other side, and A_j has endpoints y_{j-1} and y_{j-3}. Then

$$f(A_j) \supseteq \text{segment between } f(y_{j-1}) = y_j \text{ and } f(y_{j-3}) = y_{j-2}.$$

Since y_j lies beyond y_{j-1} on the side of y_{j-1}, this segment contains $B_{j-1} \cup A_{j+1} = B_j$. In particular $A_j \to A_i$ for every $i \le j+1$. For $j \ge 3$ the loop $A_{j-2} \to A_{j-1} \to A_j \to A_{j-2}$ has length 3 and is not a repetition. For $j = 2$ the loop $A_1 \to A_2 \to A_2 \to A_1$ does the same job. Since $3 < p$, Lemma B gives a period-3 orbit, a contradiction. So the sides alternate: A_{j+1} lies on the right for odd j and on the left for even j.

(d) The picture. For $p = 9$, ordering the points from left to right gives $y_6 < y_4 < y_2 < y_0 < y_{-1} < y_1 < y_3 < y_5 < y_7$, which is $x_1 < \cdots < x_9$. The partition intervals read from left to right are therefore $A_7, A_5, A_3, A_1, A_2, A_4, A_6, A_8$, as in Figure 3.15. The orbit is

$$x_5 \to x_4 \to x_6 \to x_3 \to x_7 \to x_2 \to x_8 \to x_1 \to x_9 \to x_5,$$

or its mirror image. Check the endpoint images: for each $i \le 7$, A_i has endpoints y_{i-1}, y_{i-3} and maps to the endpoints y_i, y_{i-2} of A_{i+1}, so $f(A_i) \supseteq A_{i+1}$. For the last interval, $A_8 = [x_8, x_9]$ maps to the segment between $f(x_8) = x_1$ and $f(x_9) = x_5$, which is $[x_1, x_5] = A_7 \cup A_5 \cup A_3 \cup A_1$. Hence

$$A_1 \subseteq f(A_8), \quad \text{and} \quad A_8 \to A_i \text{ for every odd } i.$$

The transition graph (Figure 3.16) has the arrows $A_1 \to A_1$, $A_i \to A_{i+1}$ for $1 \le i \le 7$, and $A_8 \to A_1, A_3, A_5, A_7$.

Step 6: Periodic points for $p = 9$

Every loop below uses only arrows listed in Step 5(d). Extra arrows do not matter.

(a) Even periods less than 9. For odd i the loop $A_8 \to A_i \to A_{i+1} \to \cdots \to A_8$ has length $1 + (8 - i) = 9 - i$. Taking $i = 7, 5, 3, 1$ gives lengths $2, 4, 6, 8$. Each loop has no repeated interval, so it is not a repetition, and $9 \nmid n$, so Lemma B gives orbits of least period $2, 4, 6, 8$.

(c) Period 1. $A_1 \to A_1$ means $f(A_1) \supseteq A_1$, so f has a fixed point in A_1 by Theorem 3.17. It is not in P, which has no fixed points.

(b) Every period $n > 9$. Let $r = n - 7 \geq 3$ and use the loop

$$\underbrace{A_1 \to \cdots \to A_1}_{r} \to A_2 \to A_3 \to \cdots \to A_8 \to A_1$$

of length $r + 7 = n$. It has a single run of A_1's, so it is not a repetition. If the resulting point x were in P, then $f^t(x) \in A_1 \cap P = \{u, w\}$ for $t = 0, \ldots, r - 1$. But $f(u) = y_1 \notin A_1$ and $f(w) = u$, so an orbit point can stay in A_1 for at most two consecutive steps, a contradiction since $r \geq 3$. By Lemma B, x has least period n. (This also covers multiples of 9.)

Hence f has orbits of period 1, all even periods less than 9, and all periods greater than 9.

Step 7: Any smallest odd period $p > 1$

Let p be the smallest odd period larger than 1.

Case $p = 3$. Step 4 is not needed. Write $P = \{x_1 < x_2 < x_3\}$ and, after reflecting if necessary, $x_1 \to x_2 \to x_3 \to x_1$. Then $A_1 = [x_2, x_3]$ and $A_2 = [x_1, x_2]$, with $f(A_1) \supseteq [x_1, x_3]$ and $f(A_2) \supseteq [x_2, x_3] = A_1$. The graph is $A_1 \to A_1, A_2$ and $A_2 \to A_1$, which is the pattern of Step 5(d).

Case $p \geq 5$. Steps 3 to 5 used only that p is odd and that f has no orbit of period 3 or $p - 2$, so they hold for any such p. The same cycle and alternation give y_j on the right for odd j and on the left for even j, so $A_{p-1} = [y_{p-4}, y_{p-2}]$ and

$$f(A_{p-1}) \supseteq \text{segment from } f(y_{p-2}) = w \text{ to } f(y_{p-4}) = y_{p-3} \ ,$$

and y_{p-3} is the extreme left point. This segment contains A_1 and every interval on the left, which are the A_i with i odd. So $A_{p-1} \to A_i$ for all odd $i < p$.

The loops of Step 6 carry over. The loop $A_{p-1} \to A_i \to \cdots \to A_{p-1}$ with i odd has length $p - i$, which runs through all even numbers from 2 to $p - 1$. The loop $A_1 \to A_1$ gives period 1. For $n > p$ use $r = n - (p - 2) \geq 3$ copies of A_1 followed by $A_2, \ldots, A_{p-1}$, with the same argument that a point of P cannot visit A_1 three times in a row. So f has orbits of all periods except possibly the odd numbers strictly between 1 and p, which is Sharkovskii's theorem for a smallest odd period p.

Step 8: An even period forces a period-two orbit

Let p be the smallest even period of f. If $p = 2$ there is nothing to prove, so assume $p \geq 4$ and take a period-p orbit P. Apply Step 4. Alternative (1) would give an even period $p - 2 < p$, contradicting minimality. Alternative (2) gives the period-two orbit directly. So assume $k = p - 2$. Steps 5(a) and 5(b) are unaffected by parity, so the cycle $w \to u \to y_1 \to \cdots \to y_{p-2} \to w$ and the arrows $A_i \to A_{i+1}$ are available.

If the sides s_j stop alternating at some $j \geq 2$, then as in Step 5(c) $f(A_j) \supseteq B_j \supseteq A_{j-1}$, and $A_{j-1} \to A_j \to A_{j-1}$ is a loop of length 2. It gives a point of least period 2, because $p \neq 2$ keeps it out of P.

If the sides alternate throughout, then for even p the extreme points are y_{p-2} (left) and y_{p-3} (right), and $A_{p-1} = [y_{p-4}, y_{p-2}]$ maps over the segment from $f(y_{p-2}) = w$ to $f(y_{p-4}) = y_{p-3}$. This segment contains A_{p-2}, the interval with outer endpoint y_{p-3}. So $A_{p-1} \to A_{p-2} \to A_{p-1}$ is a loop of length 2, and again there is a period-two orbit.

In every case f has a period-two orbit.

Step 9: Periods under f^{2^k}

Let $g = f^{2^k}$ and let x have least period r under g. Then x is f-periodic, say of least period m. The point satisfies $g^i(x) = x$ exactly when $m \mid 2^k i$, so $r = m/\gcd(m, 2^k)$. Write $\gcd(m, 2^k) = 2^j$ with $0 \le j \le k$. Then

$$m = 2^j r,$$

so x has period $2^j r$ under f for some $0 \le j \le k$.

If r is even, then $v_2(m) = v_2(r) + j > j$, where v_2 is the exponent of 2. But $2^j = \gcd(m, 2^k)$ means $j = \min(v_2(m), k)$. Since $j < v_2(m)$, we must have $j = k$. So an even r forces $m = 2^k r$.

Step 10: Period 2^k forces all smaller powers of two

Let f have a point of period 2^k with $k \ge 2$ and put $h = f^{2^{k-2}}$. By the formula in Step 9 this point has least period $2^k/2^{k-2} = 4$ under h. By Step 8, h has a point z of least period 2. By Step 9 with $r = 2$ even, z has least period $2^{k-2} \cdot 2 = 2^{k-1}$ under f. So a period-2^k orbit gives a period-2^{k-1} orbit. Repeating, f has orbits of periods $2^{k-1}, \ldots, 4, 2$. Finally a period-two orbit gives a fixed point, by the argument given at the start of the challenge ($f([x_1, x_2]) \supseteq [x_1, x_2]$ and Theorem 3.17).

Step 11: Assembling the theorem

Let f have a periodic orbit of period p. We show it has orbits of every period q with $p \rhd q$.

If p is a power of 2, Step 10 gives all smaller powers of two, which are all numbers to the right of p.

Otherwise, the numbers that are not powers of two are well ordered by the Sharkovskii ordering (first by the power of two, then by the odd part). So among the periods of f that are not powers of two there is a leftmost one, $p' = 2^k q$ with $q > 1$ odd, and $p' \rhd p$ or $p' = p$. It suffices to treat p'.

Let $g = f^{2^k}$. A point of period $2^k q$ for f has period q under g (Step 9's formula). If g had an odd period q' with $1 < q' < q$, then Step 9 would give f a period $2^j q'$ with $j \le k$, which lies to the left of $2^k q$ and contradicts the choice of p'. So q is the smallest odd period larger than 1 of g, and Step 7 applies to g. It gives g orbits of all even periods r, of all periods $r > q$, and of period 1.

- For every even r, Step 9 gives f an orbit of period exactly $2^k r$. These include $2^{k'} \cdot \text{odd}$ for $k' > k$ and the powers $2^{k'}$ for $k' > k$.
- For odd $r > q$, Step 9 gives f a period $2^j r$ with $j \le k$. If $j < k$ this would lie to the left of $2^k q$, which is impossible, so $j = k$ and f has period $2^k r$.

- The powers of two. f has a period-2^{k+1} orbit (the case $r = 2$), so by Step 10 it has all smaller powers of two, and by the first bullet all larger ones.

These are exactly the numbers to the right of $2^k q$ in the Sharkovskii ordering. □

4 Challenge 4: Fractal Basin Boundaries and the Uncertainty Exponent

Setting. The map is

$$F(x) = \begin{cases} 5x + 4, & x \le -0.4, \\ -5x, & -0.4 \le x \le 0.4, \\ 5x - 4, & x \ge 0.4, \end{cases}$$

which is continuous (the pieces agree at $x = \pm 0.4$, where $F = \mp 2$). It has fixed points -1, 0 and 1. Let $B_{+\infty}$ and $B_{-\infty}$ be the sets of points whose orbits tend to $+\infty$ and $-\infty$, and let J be the set of points whose orbits stay in $[-1, 1]$ forever.

Strategy. Steps 1 to 4 identify J as a Cantor set that is the common boundary of the two basins. Steps 5 to 8 show that the length (or area) of ε-uncertain points scales like ε^p, where p is the codimension of the boundary.

Step 1: Describing $B_{+\infty}$, $B_{-\infty}$ and J

Cut $[-1, 1]$ into five equal parts of length 0.4 and track each under F.

piece	formula for F	image
$[-1, -0.6]$	$5x + 4$ (increasing)	$[-1, 1]$
$[-0.6, -0.2]$	$5x + 4$ on $[-0.6, -0.4]$, $-5x$ on $[-0.4, -0.2]$	$[1, 2]$
$[-0.2, 0.2]$	$-5x$ (decreasing)	$[-1, 1]$
$[0.2, 0.6]$	$-5x$ on $[0.2, 0.4]$, $5x - 4$ on $[0.4, 0.6]$	$[-2, -1]$
$[0.6, 1]$	$5x - 4$ (increasing)	$[-1, 1]$

For $y > 1$ we have $F(y) = 5y - 4 > y$, so the orbit increases without bound. Hence $x > 1$ and every $x \in (-0.6, -0.2)$ (which maps into $(1, 2]$) belong to $B_{+\infty}$. Symmetrically, $F(y) = 5y + 4 < y$ for $y < -1$, so $x < -1$ and every $x \in (0.2, 0.6)$ belong to $B_{-\infty}$.

The three remaining pieces each map affinely onto all of $[-1, 1]$, so the same cut can be repeated inside each. At stage n the points of $[-1, 1]$ whose first n iterates stay in $[-1, 1]$ form 3^n closed intervals of length $2 \cdot 5^{-n}$, and the second and fourth fifths of each are removed at the next stage. Therefore

$$J = \bigcap_{n \ge 0} J_n, \qquad J_n = \{x : F^j(x) \in [-1, 1] \text{ for } 0 \le j \le n\},$$

is a Cantor set. In the coordinate $s = (x + 1)/2 \in [0, 1]$ the map becomes $s \mapsto 5s$, $s \mapsto 3 - 5s$, $s \mapsto 5s - 4$ on the three outer pieces, and J corresponds to the numbers $s \in [0, 1]$ whose base-5 expansion uses only the digits $0, 2, 4$.

If $x \notin J$ the orbit eventually leaves $[-1,1]$. At the first exit time n the point $F^{n-1}(x)$ lies in $[-1,1]$ and $F^n(x) \notin [-1,1]$, which means $F^{n-1}(x) \in (-0.6,-0.2) \cup (0.2,0.6)$, and the orbit then tends to $+\infty$ or $-\infty$ as shown above. So

$$B_{+\infty} = \{x : F^n(x) > 1 \text{ for some } n\}, \qquad B_{-\infty} = \{x : F^n(x) < -1 \text{ for some } n\},$$

and $\mathbb{R} = B_{+\infty} \sqcup B_{-\infty} \sqcup J$. Each basin is the union of $(1,\infty)$, respectively $(-\infty,-1)$, and infinitely many open intervals removed at the various stages of the construction.

Step 2: $\operatorname{boxdim}(J) = \ln 3 / \ln 5$

Let $\varepsilon_n = 5^{-n}$. The set J lies in the 3^n stage-n intervals, each of length $2\varepsilon_n$ and each covered by at most 3 boxes of side ε_n. So $N(\varepsilon_n) \le 3 \cdot 3^n$. Conversely, the gaps between stage-n intervals have length at least $2 \cdot 5^{-n} = 2\varepsilon_n$, so a box of side ε_n meets at most one of them, and each of them contains points of J (its endpoints map to the fixed points ± 1). So $N(\varepsilon_n) \ge 3^n$. Therefore

$$\frac{\ln N(\varepsilon_n)}{\ln(1/\varepsilon_n)} \in \left[\frac{n \ln 3}{n \ln 5}, \frac{n \ln 3 + \ln 3}{n \ln 5}\right] \longrightarrow \frac{\ln 3}{\ln 5}.$$

For ε between $5^{-(n+1)}$ and 5^{-n} the number $N(\varepsilon)$ lies between $N(5^{-n})$ and $N(5^{-(n+1)})$, which gives the same limit. Hence $\operatorname{boxdim}(J) = \ln 3 / \ln 5 \approx 0.683$.

Step 3: The basins are open

Let $x \in B_{+\infty}$, so $F^n(x) > 1$ for some n. F^n is continuous, so there is $\varepsilon > 0$ with $F^n(y) > 1$ for all $y \in N_\varepsilon(x)$. Each such y then belongs to $B_{+\infty}$, so $N_\varepsilon(x) \subseteq B_{+\infty}$. The argument for $B_{-\infty}$ is identical.

Step 4: Every point of J borders both basins

Let $y \in J$ and $\varepsilon > 0$. Choose n with $2 \cdot 5^{-n} < \varepsilon$ and let I be the stage-n interval containing y. Then $I \subseteq N_\varepsilon(y)$. Inductively, F^n maps I affinely and bijectively onto $[-1,1]$, because at each step the image lies in one of the three outer pieces, each of which F maps affinely onto $[-1,1]$. Hence I contains a point z_1 with $F^n(z_1) = -0.4$ and a point z_2 with $F^n(z_2) = 0.4$. Since $F(-0.4) = 2 > 1$ and $F(0.4) = -2 < -1$, we get $z_1 \in B_{+\infty}$ and $z_2 \in B_{-\infty}$. So $N_\varepsilon(y)$ meets both basins.

Together with Step 3 this shows $J = \partial B_{+\infty} = \partial B_{-\infty}$. A boundary point of an open basin cannot lie in that basin, and it cannot lie in the other open basin, which is disjoint from a neighborhood of it. So it lies in J, and Step 4 gives the reverse inclusion.

Step 5: $\varepsilon N(\varepsilon) \le L(\varepsilon) \le 3\varepsilon N(\varepsilon)$

Fix a grid of intervals of length ε. Let $N(\varepsilon)$ be the number of grid intervals that meet the boundary J. The ε-uncertain points are the points within distance ε of J, and $L(\varepsilon)$ is their total length.

Lower bound. Every point of a grid interval that contains a point $y \in J$ is within ε of y, so it is ε-uncertain. The $N(\varepsilon)$ grid intervals overlap only at endpoints, so $L(\varepsilon) \ge N(\varepsilon)\,\varepsilon$.

Upper bound. If x is within ε of $y \in J$ and y lies in the grid interval $[a, a+\varepsilon]$, then $x \in (a-\varepsilon, a+2\varepsilon)$, an interval of length 3ε. These $N(\varepsilon)$ intervals cover all uncertain points, so $L(\varepsilon) \le 3\varepsilon N(\varepsilon)$.

Check. For $\varepsilon = 0.1$ the uncertain points in $[-1,1]$ are $(-1,-0.5)$, $(-0.3,0.3)$ and $(0.5,1)$, of total length 1.6.

Step 6: The uncertainty exponent $p = 1 - \mathrm{boxdim}(J)$

Take logarithms in Step 5, $\ln\varepsilon + \ln N(\varepsilon) \le \ln L(\varepsilon) \le \ln 3 + \ln\varepsilon + \ln N(\varepsilon)$, and divide by $\ln\varepsilon < 0$, which reverses the inequalities,

$$1 + \frac{\ln N(\varepsilon)}{\ln\varepsilon} \;\ge\; \frac{\ln L(\varepsilon)}{\ln\varepsilon} \;\ge\; 1 + \frac{\ln N(\varepsilon)}{\ln\varepsilon} + \frac{\ln 3}{\ln\varepsilon}.$$

If $d = \mathrm{boxdim}(J) = \lim_{\varepsilon\to 0} \ln N(\varepsilon)/\ln(1/\varepsilon)$ exists, then $\ln N/\ln\varepsilon \to -d$ and $\ln 3/\ln\varepsilon \to 0$. Both bounds tend to $1-d$, so

$$\lim_{\varepsilon\to 0} \frac{\ln L(\varepsilon)}{\ln\varepsilon} = 1 - d = p, \qquad \text{that is} \quad L(\varepsilon) \approx \varepsilon^p.$$

For the map F this gives $p = 1 - \ln 3/\ln 5 \approx 0.317$.

Step 7: A reduction of L by a factor of two

With $\mathrm{boxdim}(J) = 0.9$ we have $p = 0.1$ and $L(\varepsilon) \approx C\varepsilon^{0.1}$. To halve L we need $(\varepsilon'/\varepsilon)^{0.1} = \frac{1}{2}$, so

$$\frac{\varepsilon'}{\varepsilon} = 2^{-10} = \frac{1}{1024}.$$

The measurement accuracy has to improve by a factor of 1024, about three decimal digits, to cut the uncertain length in half. A larger dimension of J means a smaller p and a much weaker payoff from better measurements.

Step 8: The two-dimensional version

Let $J \subset \mathbb{R}^2$ be a basin boundary each of whose points lies in the boundary of at least two basins. This hypothesis means every point within ε of J has a point of at least two different basins within ε of it, so such points are ε-uncertain. Let $A(\varepsilon)$ be the area of the set of points within ε of J, and $N(\delta)$ the number of grid squares of side δ that meet J.

Upper bound. If x is within ε of $y \in J$ and y lies in a grid square S of side ε, then x lies in the concentric square of side 3ε around S. So $A(\varepsilon) \le 9\varepsilon^2 N(\varepsilon)$.

Lower bound. Use squares of side $\delta = \varepsilon/\sqrt{2}$. Each point of such a square is within its diameter $\delta\sqrt{2} = \varepsilon$ of the point of J it contains, so the squares lie in the uncertain set, and $A(\varepsilon) \ge \delta^2 N(\delta) = \frac{1}{2}\varepsilon^2 N(\varepsilon/\sqrt{2})$.

Take logarithms and divide by $\ln\varepsilon < 0$, exactly as in Step 6. The ratios $\ln N(\varepsilon/\sqrt{2})/\ln(1/\varepsilon)$ and $\ln N(\varepsilon)/\ln(1/\varepsilon)$ have the same limit $d = \mathrm{boxdim}(J)$, so

$$\lim_{\varepsilon\to 0} \frac{\ln A(\varepsilon)}{\ln\varepsilon} = 2 - d = p,$$

so $A(\varepsilon) \approx C\varepsilon^p$ with $p = 2 - \mathrm{boxdim}(J)$. The constant 2 here is the dimension of the ambient space, so in general p is the codimension of the boundary. □

5 Challenge 5: Computer Calculations and Shadowing

Setting. The skinny baker map on the unit square is

$$B(x_1, x_2) = \begin{cases} \left(\frac{1}{3}x_1,\ 2x_2\right), & 0 \le x_2 \le \frac{1}{2}, \\ \left(\frac{1}{3}x_1 + \frac{2}{3},\ 2x_2 - 1\right), & \frac{1}{2} < x_2 \le 1. \end{cases}$$

On each half it is affine, contracting horizontally by $\frac{1}{3}$ and stretching vertically by 2. Its fixed points are $(0,0)$ and $(1,1)$. The cat map is $S(v) = Av \pmod 1$ with $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

Tools from the text. Say that $f(A)$ *lies across* B *vertically* when the images of the left and right sides of A lie in the vertical strip bounded by the sides of B, while the image of the top side lies on or above the top of B and the image of the bottom side lies on or below the bottom of B. The Colorado Corollary (Corollary 5.12) says that if $f(S)$ lies across S then f has a fixed point in S. Lying across is transitive (Figure 5.10), and Corollary 5.13 says that a closed chain $f(S_1)$ across $S_2, \ldots, f(S_k)$ across S_1 gives a fixed point of f^k in S_1.

Strategy. A computed orbit is a "pseudo-orbit" in which each point is only close to the image of the previous one. For the baker map each step expands the error in one direction and contracts it in the other. Surrounding each computed point by a suitably shaped rectangle makes the images lie across the next rectangles, and the fixed-point machinery produces a true orbit nearby.

Step 1: A small gap forces a fixed point nearby

Let $x_0 = (a, b)$ and $B(x_0) = (a', b')$ with $|a' - a| < d$ and $|b' - b| < d$. Let $S_0 = [a - \frac{3}{2}d, a + \frac{3}{2}d] \times [b - d, b + d]$, the $3d \times 2d$ rectangle centered at x_0, and assume it lies on one side of $x_2 = \frac{1}{2}$. Then on S_0 the map B is the affine map $x \mapsto (\frac{1}{3}x_1 + c,\ 2x_2 - e)$ for fixed constants (c, e). The image $B(S_0)$ is the rectangle centered at $B(x_0)$ of width $\frac{1}{3} \cdot 3d = d$ and height $2 \cdot 2d = 4d$.

Horizontal sides. The image occupies $[a' - \frac{d}{2},\ a' + \frac{d}{2}]$ in the x_1 direction. Since $|a' - a| < d$,

$$a' + \tfrac{d}{2} < a + \tfrac{3}{2}d, \qquad a' - \tfrac{d}{2} > a - \tfrac{3}{2}d,$$

so the images of the left and right sides lie strictly inside the strip $a - \frac{3}{2}d < x_1 < a + \frac{3}{2}d$.

Vertical sides. The image occupies $[b' - 2d,\ b' + 2d]$ in the x_2 direction. Since $|b' - b| < d$,

$$b' + 2d > b + d, \qquad b' - 2d < b - d,$$

so the image of the top side lies above the top of S_0 and the image of the bottom side lies below the bottom.

Hence $B(S_0)$ lies across S_0 vertically. The Colorado Corollary, applied to the affine extension of $B|_{S_0}$ to the plane, gives a fixed point $z \in S_0$. It is the fixed point of the affine piece, namely $(0,0)$ or $(1,1)$, so it is a genuine fixed point of B. Every point of S_0 is within $\sqrt{(3d/2)^2 + d^2} = \frac{\sqrt{13}}{2}\, d < 2d$ of x_0. So B has a fixed point within $2d$ of x_0.

Step 2: Rounding errors of size 10^{-6}

Suppose the computed value $\hat{B}(x_0)$ satisfies $|\hat{B}(x_0) - B(x_0)| \le 10^{-6}$ in each coordinate and the computer reports $|\hat{B}(x_0) - x_0| \le 10^{-6}$ in each coordinate. Then each coordinate of $B(x_0) - x_0$ has absolute value at most $2 \cdot 10^{-6}$, which is less than $d = 3 \cdot 10^{-6}$. If x_0 is near a point where B is nearly the identity, then either $x_2 < d$ (lower piece, where $B(x)_2 - x_2 = x_2$) or $x_2 > 1 - d$ (upper piece), so the rectangle S_0 does not meet the line $x_2 = \frac{1}{2}$. Step 1 applies with this d, so B has a fixed point within $2d = 6 \cdot 10^{-6} < 10^{-5}$ of x_0 (in fact within $\frac{\sqrt{13}}{2} d \approx 5.4 \cdot 10^{-6}$).

Step 3: Theorem 5.19

Let $x_0, \dots, x_{k-1}, x_k = x_0$ be points of the unit square such that each coordinate of $B(x_i) - x_{i+1}$ has absolute value less than d. Let S_i be the $3d \times 2d$ rectangle centered at x_i (see the remark below for rectangles that stick out of the square or meet $x_2 = \frac{1}{2}$). Exactly the computation of Step 1, with x_{i+1} in place of x_i as the center of the target, shows that $B(S_i)$ lies across S_{i+1} vertically:

$$B(S_i) \text{ has width } d, \text{ height } 4d, \text{ center } B(x_i), \qquad |B(x_i) - x_{i+1}|_{\text{each coordinate}} < d.$$

So the horizontal extent of $B(S_i)$ is strictly inside that of S_{i+1}, and its vertical extent strictly contains that of S_{i+1}. All the rectangles have the same (vertical) orientation, and $S_k = S_0$. By Corollary 5.13, B^k has a fixed point $z_0 \in S_0$ with $B^i(z_0) \in S_i$ for $0 \le i < k$. Put $z_i = B^i(z_0)$. Then $\{z_0, \dots, z_{k-1}\}$ is a periodic orbit (of period dividing k), and since $z_i \in S_i$,

$$|x_i - z_i| \le \tfrac{\sqrt{13}}{2}\, d < 2d.$$

Remark on boundaries. B is discontinuous across $x_2 = \frac{1}{2}$, and a rectangle S_i may stick out of the unit square. Let F_i be the affine piece that agrees with B at x_i. Replace S_i by $S_i \cap [0,1]^2 \cap \{\text{the closed half of the square on the side of } x_i\}$, so that $B = F_i$ on S_i and any point found in S_i obeys the true map. The inequalities above survive. A clipped side is mapped exactly onto a side of the unit square (the top at $x_2 = \frac{1}{2}$ goes to $x_2 = 1$, the bottom at $x_2 = \frac{1}{2}$ goes to $x_2 = 0$, and the edges $x_1 = 0, 1$ go to $x_1 = c$, $c + \frac{1}{3}$ with $c \in \{0, \frac{2}{3}\}$). A target rectangle that has been clipped lies in $[0,1]^2$, so it is still inside the image. The definition of lying across allows the image to lie *on* the top or bottom of the target. So all the hypotheses of the corollaries hold, and the clipped rectangles are nondegenerate because each contains a segment of length at least d on a side of x_i.

Step 4: A chain of rectangles gives a point following the chain

Let f be continuous and let $S_0, \dots, S_k$ be rectangles with $f(S_i)$ lying across S_{i+1}, all with the same orientation (say vertical).

Claim. There is $x_0 \in S_0$ with $f^i(x_0) \in S_i$ for $0 \le i \le k$.

Proof. Let $T_i = \{x \in S_0 : \ f^j(x) \in S_j \text{ for } 0 \le j \le i\}$. By transitivity of lying across (Figure 5.10), in the form used in the text, there is a closed set $V \subseteq S_0$ with $f^j(V) \subseteq S_j$ for $j \le i$ and $f^i(V)$ a curvilinear rectangle that lies across S_i. Inductively this is built as follows. The part of $f(S_0)$ inside S_1 is a curvilinear vertical strip running from the top of S_1 to its bottom, because $f(S_0)$ lies across S_1. This strip is mapped by f across S_2, because $f(S_1)$ lies across S_2, and the portion in S_2 is again a strip spanning S_2, and so on. In particular $T_i \ne \varnothing$ for

every $i \le k$, and T_k is the set of points that follow the chain. Each T_i is closed, since f is continuous and the S_j are closed. □

Must k be finite? No. The sets $T_0 \supseteq T_1 \supseteq T_2 \supseteq \cdots$ are nonempty, closed and nested inside the compact rectangle S_0, so by the finite intersection property $\bigcap_i T_i \neq \varnothing$. A point of the intersection satisfies $f^i(x_0) \in S_i$ for all $i \ge 0$.

Step 5: True orbits near arbitrary computed orbits of the baker map

Let $x_0, \ldots, x_k$ be points with every coordinate of $B(x_i) - x_{i+1}$ smaller than d in absolute value, for $0 \le i < k$. As in Step 3, the rectangles S_i satisfy: $B(S_i)$ lies across S_{i+1} vertically. Step 4 gives $z_0 \in S_0$ with $B^i(z_0) \in S_i$ for $0 \le i \le k$. Setting $z_i = B^i(z_0)$ we get a true orbit with

$$|x_i - z_i| < 2d \qquad (0 \le i \le k).$$

No periodicity is needed. This is the shadowing statement, and the length k of the computed orbit does not matter.

Step 6: The cat map

The matrix A is symmetric with eigenvalues $\lambda = \frac{3+\sqrt{5}}{2} = \varphi^2$ and $\mu = \frac{1}{\lambda} = \frac{3-\sqrt{5}}{2}$, where $\varphi = \frac{1+\sqrt{5}}{2}$. The eigenvectors $e_u \parallel (1, \varphi^{-1})$ (expanding) and $e_s \parallel (-\varphi^{-1}, 1)$ (contracting) are orthogonal. Replace the $3d \times 2d$ rectangles by rectangles with sides along the eigendirections.

Lift the pseudo-orbit to $\mathbb{R}^2$ so that $\tilde{x}_{i+1}$ is the lift nearest $A\tilde{x}_i$, and suppose $|A\tilde{x}_i - \tilde{x}_{i+1}| < \delta$. Let

$$R_i = \{\tilde{x}_i + s\, e_s + t\, e_u : \ |s| \le \varphi\delta, \ |t| \le \delta/\varphi\}.$$

Then $A(R_i)$ is the rectangle centered at $A\tilde{x}_i$ with half-sizes $\mu\varphi\delta$ along e_s and $\lambda\delta/\varphi$ along e_u. Write $A\tilde{x}_i - \tilde{x}_{i+1} = c_s e_s + c_u e_u$, so that $|c_s|, |c_u| < \delta$.

- In the e_s direction, $|c_s| + \mu\varphi\delta < \delta + \mu\varphi\delta = \varphi\delta$ because $\varphi(1-\mu) = 1$. So the sides of $A(R_i)$ along e_u lie strictly inside R_{i+1}.
- In the e_u direction we need $\lambda\delta/\varphi - |c_u| \ge \delta/\varphi$, which is equivalent to $|c_u| \le (\lambda-1)\delta/\varphi = \delta$ because $\lambda - 1 = \varphi$. This holds since $|c_u| < \delta$. So the ends of $A(R_i)$ extend beyond both ends of R_{i+1}.

Hence $A(R_i)$ lies across R_{i+1} (with the expanding direction playing the role of "vertical"). The map is linear on $\mathbb{R}^2$, so there is no discontinuity to worry about, and Step 4 gives $\tilde{z}_0 \in R_0$ with $A^i\tilde{z}_0 \in R_i$. Projecting mod 1 gives a true orbit of S, because $S^i(v) = A^i v \pmod 1$. Every point of R_i is within the half-diagonal

$$\sqrt{\varphi^2 + \varphi^{-2}}\ \delta = \sqrt{3}\,\delta$$

of $\tilde{x}_i$, using $\varphi^2 + \varphi^{-2} = (\varphi + \varphi^{-1})^2 - 2 = 5 - 2 = 3$. If the pseudo-orbit is closed, $x_k = x_0$ on the torus, then $\tilde{x}_k = \tilde{x}_0 + m$ for an integer vector m, and the map $w \mapsto A^k w - m$ maps R_0 across itself, so it has a fixed point, which is a periodic orbit of the cat map.

If the hypothesis is stated coordinatewise, each coordinate of $A\tilde{x}_i - \tilde{x}_{i+1}$ smaller than d, then $\delta = \sqrt{2}\,d$ and the shadowing distance is $\sqrt{6}\,d \approx 2.45\,d$.

What matters. The proof uses three things. There is a direction in which the map expands by a factor bounded away from 1, a complementary direction in which it contracts, and these two

factors are uniform over the whole phase space, so a single choice of rectangle shape works at every step. Any map with this uniform stretching and shrinking (a hyperbolic structure) can be treated the same way, and the Postscript lists several that have been.

Step 7: A computed cat-map orbit of length 10^6

If each iteration is computed to within 10^{-6} then $\delta \approx 10^{-6}$ (or $\sqrt{2} \cdot 10^{-6}$ if the bound is coordinatewise). Step 6 gives a true orbit within $\sqrt{3} \cdot 10^{-6} \approx 1.7 \cdot 10^{-6}$ (respectively $2.5 \cdot 10^{-6}$) of every one of the 10^6 plotted points. A pixel on a screen is about 10^{-3} of the unit square, a thousand times larger. So yes, the plotted dots represent a true orbit to within pixel resolution. The true orbit that stays close may start at a slightly different initial condition from the one typed in. It exists regardless, so the picture is faithful to the dynamics.

Step 8: What map (5.14) lacks

The map $f(x, y) = (x + d, y + d)$ is a translation. Its derivative is the identity, so it neither stretches nor contracts. A rectangle is carried to a congruent rectangle shifted by (d, d), so $f(S)$ never lies across S in the required sense, and an error of size d at each step is simply carried along and accumulates linearly. In the baker and cat maps every step stretches errors in one direction and shrinks them in the other, which is what allows a nearby true orbit to absorb the error. The missing property is hyperbolicity: uniform stretching in some direction (equivalently, the presence of both positive and negative Lyapunov exponents), together with the compensating contraction.

11 Challenge 11: Hamiltonian Systems and the Lyapunov Center Theorem

Notation. Write $v = (x, y)$, $\nabla H = (H_x, H_y)^T$, and

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J^T = -J, \qquad J^2 = -I, \qquad \det J = 1.$$

Then (11.19) reads $\dot{v} = J\nabla H(v)$. Let M denote the 2×2 Hessian matrix of H at the origin.

Strategy. The Hamiltonian system (11.19) is embedded in the family (11.20). That family has a Hopf bifurcation at $a = 0$ and no periodic orbits for $a \neq 0$. Every periodic orbit supplied by the Hopf theorem must therefore belong to the original system at $a = 0$.

Step 1: H is constant along solutions

Along a solution of (11.19),

$$\dot{H} = H_x\dot{x} + H_y\dot{y} = H_xH_y + H_y(-H_x) = 0.$$

Hence H is a constant of the motion.

Step 2: $\{(a,0)\}$ is a path of equilibria

The origin is an equilibrium of (11.19), so $H_y(0) = H_x(0) = 0$, that is $\nabla H(0) = 0$. The family (11.20) can be written $\dot{v} = (aI + J)\nabla H(v)$, and at $v = 0$ the right-hand side equals $(aI + J)\,0 = 0$ for every a. So $\{(a,0) : a \in \mathbb{R}\}$ is a path of equilibria.

Step 3: No periodic orbits for $a \neq 0$

Let $v(t)$ solve (11.20). By the chain rule,

$$\frac{dH}{dt} = \nabla H \cdot \dot{v} = \nabla H^T (aI + J)\nabla H = a\,|\nabla H|^2 + \nabla H^T J\,\nabla H = a\,|\nabla H|^2,$$

because J is antisymmetric and so $w^T J w = 0$ for every vector w.

Suppose $a \neq 0$ and the solution is a nonconstant periodic orbit of period T. Then $H(v(T)) - H(v(0)) = 0$, which gives

$$0 = a \int_0^T |\nabla H(v(t))|^2\,dt.$$

The orbit is not an equilibrium, so $\nabla H \neq 0$ at every point of it (where $\nabla H = 0$ the vector field vanishes). The integrand is then strictly positive and the integral is positive, a contradiction. Thus (11.20) has no nonequilibrium periodic orbits when $a \neq 0$.

Step 4: M and M^{-1} are symmetric, and M is nonsingular

Symmetric. H is smooth, so mixed partials agree, $H_{xy} = H_{yx}$, and $M = M^T$.

Nonsingular. The Jacobian of (11.19) at 0 is $A = JM$, with eigenvalues $\pm bi$ and $b \neq 0$. Therefore $\det A = (bi)(-bi) = b^2 > 0$. Since $\det J = 1$, we get $\det M = b^2 \neq 0$.

Inverse. $(M^{-1})^T = (M^T)^{-1} = M^{-1}$.

Step 5: $\dot{v} = JMv$ is Hamiltonian with $E(v) = \frac{1}{2}v^T M v$

Because M is symmetric, $\nabla E(v) = \frac{1}{2}(M + M^T)v = Mv$. Hamilton's equations for E are $\dot{v} = J\nabla E(v) = JMv$, which is (11.21). This is also the linearization of (11.19) at 0, since $D(J\nabla H)(0) = JM$.

Step 6: If $a \neq 0$ then $c(a) \neq 0$

Take $|a|$ small, so that $(aI + J)M$ has a complex conjugate pair of eigenvalues $c(a) \pm d(a)i$ near $\pm bi$. Suppose $c(a) = 0$ for some $a \neq 0$.

First, $d(a) \neq 0$. Otherwise 0 would be an eigenvalue, yet $\det[(aI + J)M] = (a^2 + 1)\det M \neq 0$ by Step 4.

The matrix therefore has pure imaginary eigenvalues $\pm d(a)i$ with $d(a) \neq 0$, and the linear system $\dot{v} = (aI + J)Mv$ has a linear center at the origin. All its nonzero solutions are periodic with period $2\pi/|d(a)|$.

This linear system equals $\dot{v} = (aI + J)\nabla E(v)$, which is exactly the form (11.20) with Hamiltonian E in place of H. Step 3 applied to E says that it has no nonconstant periodic orbits when $a \neq 0$. This contradiction shows that $c(a) \neq 0$ for $a \neq 0$.

Step 7: λ an eigenvalue of $(aI+J)M$ implies $-\lambda$ is an eigenvalue of $(-aI+J)M$

Suppose $(aI + J)M - \lambda I$ is singular. Its transpose is singular too, and using $M^T = M$, $J^T = -J$,

$$\big[(aI + J)M - \lambda I\big]^T = M(aI - J) - \lambda I.$$

Multiplying on the left by M^{-1} and on the right by M keeps the matrix singular, and

$$M^{-1}\big[M(aI - J) - \lambda I\big]M = (aI - J)M - \lambda I = -\big[(-aI + J)M - (-\lambda)I\big].$$

So $(-aI + J)M - (-\lambda)I$ is singular, which says $-\lambda$ is an eigenvalue of $(-aI + J)M$.

Step 8: The eigenvalues cross the imaginary axis at $\pm bi$ when $a = 0$

At $a = 0$ the matrix is JM with eigenvalues $\pm bi$, so $c(0) = 0$ and, after labeling the continuous path, $d(0) = b \neq 0$.

Apply Step 7 to $\lambda = c(a) + d(a)i$. Then $-c(a) - d(a)i$ is an eigenvalue of $(-aI + J)M$. The eigenvalues of that matrix are $c(-a) \pm d(-a)i$, and both share the real part $c(-a)$. Comparing real parts gives

$$c(-a) = -c(a),$$

so c is an odd function. By Step 6 it is nonzero for $a \neq 0$, and by continuity it keeps one sign for $a > 0$. Oddness forces the opposite sign for $a < 0$. Hence c changes sign at $a = 0$, and the pair of eigenvalues crosses the imaginary axis at $\pm bi$.

Remark (a direct check). For a complex pair the real part is half the trace. Writing $M = \begin{pmatrix} p & q \\ q & r \end{pmatrix}$ gives $JM = \begin{pmatrix} q & r \\ -p & -q \end{pmatrix}$, so $\operatorname{tr}(JM) = 0$ and $c(a) = \frac{1}{2}a(p + r)$. Since $pr - q^2 = b^2 > 0$, the entries p and r share a sign and $p + r \neq 0$. So $c'(0) \neq 0$ and the crossing even has nonzero speed. The weak form of the Hopf theorem is all the argument needs.

Step 9: Apply the weak Hopf theorem

The family (11.20) lives in $\mathbb{R}^2$ and satisfies the hypotheses of Remark 11.22.

- $v = 0$ is an equilibrium for every a (Step 2).
- A pair $c(a) \pm d(a)i$ of eigenvalues of $D_v f_a(0)$ crosses the imaginary axis at $a = 0$, with $b \neq 0$ (Step 8).
- No other eigenvalue is an integer multiple of bi. In the plane the only other eigenvalue is the conjugate $-bi$, so there is nothing further to check.

Remark 11.22 then says that every neighborhood of $(a, v) = (0, 0)$ contains a periodic orbit of some member of the family. By Step 3 no member with $a \neq 0$ has one, so all of these orbits belong to $a = 0$, the Hamiltonian system (11.19). Their periods approach $2\pi/b$ as the orbits shrink to the origin, as in Theorem 11.21.

Every neighborhood of the center therefore contains periodic orbits whose periods approach $2\pi/b$, which is the Lyapunov Center Theorem in the plane. □

12 Challenge 12: Universality in Bifurcation Diagrams

Setup. The canonical family is $g_a(x) = x^2 - a$. In a period-n window, a_s is the saddle-node parameter, a_d the first period-doubling parameter and a_c the crisis parameter. The ratio of interest is

$$R = \frac{a_c - a_s}{a_d - a_s}.$$

The book prints the target value as "94", which is $9/4$.

Strategy. Step 1 computes R for the canonical map. Step 2 shows that a linear change of coordinates cannot alter R. Steps 3 to 6 show that after rescaling, a high iterate f^n of a generic unimodal family approaches the canonical map, so its ratio approaches $9/4$.

Step 1: $R_g = 9/4$ for the period-one window of $g_a(x) = x^2 - a$

Saddle node. Fixed points solve $x^2 - x - a = 0$, so $x = \frac{1}{2}\big(1 \pm \sqrt{1+4a}\big)$. They are born together when the discriminant vanishes, so $a_s = -\frac{1}{4}$, at $x = \frac{1}{2}$.

First period-doubling. The stable fixed point is $x_- = \frac{1}{2}\big(1 - \sqrt{1+4a}\big)$, and $g_a'(x) = 2x$. Setting $2x_- = -1$ gives $\sqrt{1+4a} = 2$, so $a_d = \frac{3}{4}$.

Crisis. Let $x_+ = \frac{1}{2}\big(1 + \sqrt{1+4a}\big)$ be the repelling fixed point, which acts as the boundary source. Since g_a is even, $g_a(-x_+) = g_a(x_+) = x_+$. The image of $[-x_+, x_+]$ is $[-a, x_+]$, because the minimum sits at the critical point 0 where $g_a(0) = -a$. The interval maps into itself exactly when $-a \geq -x_+$, that is $a \leq x_+$. Past that value points escape and the attractor is destroyed. So the crisis occurs when the critical point lands on the source, $-a = -x_+$:

$$a = \tfrac{1}{2}\big(1 + \sqrt{1+4a}\big) \iff 2a - 1 = \sqrt{1+4a} \implies 4a^2 - 8a = 0.$$

The root $a = 0$ violates $2a - 1 \geq 0$, so $a_c = 2$. At that value $x_+ = 2$ and $[-2, 2]$ maps onto itself.

Ratio.

$$R_g = \frac{a_c - a_s}{a_d - a_s} = \frac{2 + \frac{1}{4}}{\frac{3}{4} + \frac{1}{4}} = \frac{9/4}{1} = \frac{9}{4}.$$

Step 2: Linear conjugacy preserves the ratio

Write $H(a, x) = (u, y) = \big(C_1(a - a_0),\, C_2(x - x_0)\big)$. The relation $H \circ K = G \circ H$ reads, in the second coordinate,

$$C_2\big(k_a(x) - x_0\big) = g_u\big(C_2(x - x_0)\big), \qquad u = C_1(a - a_0).$$

For fixed a let $h_a(x) = C_2(x - x_0)$, a linear map. Then $k_a = h_a^{-1} \circ g_u \circ h_a$, and since the same h_a appears at every stage the conjugation passes through iteration,

$$k_a^m = h_a^{-1} \circ g_u^m \circ h_a \qquad \text{for every } m.$$

Three consequences follow.

(a) x is a period-n point of k_a exactly when $y = C_2(x - x_0)$ is a period-n point of g_u. Orbits correspond one to one.

(b) By the chain rule, $(k_a^n)'(x) = \frac{1}{C_2}\,(g_u^n)'(y)\,C_2 = (g_u^n)'(y)$. Multipliers are identical, so multiplier $+1$ (saddle node) and multiplier -1 (period-doubling, with the period-$2n$ orbit branching off) correspond. H is a linear homeomorphism of the (a,x)-plane, so it carries these curves of orbits onto each other.

(c) h_a sends the critical point of k_a to the critical point of g_u, because $k_a' = 0$ exactly when $g_u' = 0$. It also carries the attractor, its basin and the period-n source across, so the crisis, where the critical point lands on the source, corresponds as well.

Since h_a is invertible, every statement can be read in both directions, which gives the "if and only if".

Every bifurcation parameter of k is therefore the affine image $a = a_0 + u/C_1$ of the matching parameter for g, and

$$R_k = \frac{(u_c - u_s)/C_1}{(u_d - u_s)/C_1} = R_g = \frac{9}{4}.$$

The factor C_1 cancels, whatever its sign. The same cancellation applies to any ratio of distances defined by conditions on periodic points and their multipliers.

Step 3: $\partial^2 f^{n-1}/\partial x^2\,(a_0, x_1)$ is of order S^2

Let $x_1, \dots, x_{n-1}, x_n = x_0$ be the orbit. Put $L_j = \partial_x f(a_0, x_j)$, so $S = L_1 L_2 \cdots L_{n-1}$. For x near x_1 let $\xi_j(x) = f^{\,j-1}(a_0, x)$, so $\xi_1 = x$ and $\xi_k'(x) = \prod_{j<k} f_x(\xi_j)$. By the chain rule,

$$\frac{\partial f^{n-1}}{\partial x}(x) = \prod_{j=1}^{n-1} f_x\big(\xi_j(x)\big).$$

This is the product in the hint, with its factors listed from the end of the orbit back to x_1. Differentiating by the product rule,

$$\frac{\partial^2 f^{n-1}}{\partial x^2} = \sum_{k=1}^{n-1} f_{xx}(\xi_k)\,\xi_k' \prod_{j\neq k} f_x(\xi_j).$$

At $x = x_1$ set $A_k = L_1 \cdots L_{k-1}$ and $B_k = L_{k+1} \cdots L_{n-1}$, so that $S = A_k L_k B_k$. The kth term equals $f_{xx}(x_k) A_k^2 B_k = f_{xx}(x_k)\,S^2/(L_k^2 B_k)$, and therefore

$$\frac{\partial^2 f^{n-1}}{\partial x^2}(a_0, x_1) = S^2 \sum_{k=1}^{n-1} \frac{f_{xx}(x_k)}{L_k^2 B_k}.$$

Now use the working assumptions behind hypothesis (1). The orbit points $x_1, \dots, x_{n-1}$ stay far from the critical point, so f_{xx} is bounded and each $|L_j|$ is bounded away from 0. The slopes are expanding, $|L_j| \geq \lambda > 1$. Then $|B_k| \geq \lambda^{\,n-1-k}$, and the sum is dominated by a convergent geometric series whose bound does not depend on n. So the second derivative is S^2 times a bounded quantity, that is $O(S^2)$. The same pattern gives order S^k for kth order partials, including those involving a.

Step 4: The two Taylor expansions

(i) Expansion of f^{n-1} about (a_0, x_1). The base value is $f^{n-1}(a_0, x_1) = x_0$ and the x-derivative is S. For the a-derivative, a change in a is introduced at step k and then stretched by the

remaining slopes:

$$\frac{\partial f^{n-1}}{\partial a}(a_0, x_1) = \sum_{k=1}^{n-1} f_a(x_k)\, B_k = S \sum_{k=1}^{n-1} \frac{f_a(x_k)}{L_1 \cdots L_k}.$$

Let p_1 denote the last sum. Its terms shrink geometrically under the expansion assumption, so it converges as n grows, and generically its limit is nonzero. By Step 3 all second-order partials are $O(S^2)$, so Taylor's theorem gives

$$f^{n-1}(a, x) - x_0 = S(x - x_1) + p_1 S(a - a_0) + O\big(S^2(|x - x_1|^2 + |a - a_0|^2)\big).$$

The remainder is smaller than the linear terms precisely when $S^2\delta^2 \ll S\delta$, that is $\delta \ll S^{-1}$. This is the stated proviso.

(ii) Expansion of f about (a_0, x_0). Here $f(a_0, x_0) = x_1$ and $f_x(a_0, x_0) = 0$ because x_0 is the critical point. There is no linear term in x, and

$$f(a, x) - x_1 = q(x - x_0)^2 + p_2(a - a_0) + O\big(|x - x_0|^3 + |x - x_0||a - a_0| + |a - a_0|^2\big),$$

with $q = \frac{1}{2}\,\partial^2 f/\partial x^2(a_0, x_0)$ and $p_2 = \partial f/\partial a(a_0, x_0)$. The constant q is nonzero because the critical point is quadratic, and p_2 is nonzero generically.

Step 5: Composing to obtain f^n

Write $\xi = x - x_0$, $\alpha = a - a_0$ and $x' = f(a, x)$. Applying (i) with x' in place of x,

$$f^n(a, x) - x_0 = S(x' - x_1) + p_1 S\alpha + O\big(S^2(|x' - x_1|^2 + \alpha^2)\big).$$

Substitute (ii), $x' - x_1 = q\xi^2 + p_2\alpha + O(|\xi|^3 + |\xi||\alpha| + \alpha^2)$.

- The leading terms are $qS\xi^2 + (p_1 + p_2)S\alpha$. Put $p = p_1 + p_2$.
- The error inside (ii), multiplied by S, is $S \cdot O(|\xi|^3 + |\xi||\alpha| + \alpha^2)$. The α^2 piece satisfies $S\alpha^2 \le S^2\alpha^2$ because $S \ge 1$, so it is absorbed into the S^2 term below.
- Squaring $x' - x_1$ gives $O(\xi^4 + \alpha^2)$, since cross terms such as $\xi^2\alpha$ are at most $\frac{1}{2}(\xi^4 + \alpha^2)$. The error from (i) is therefore $O\big(S^2(\xi^4 + \alpha^2)\big)$.

Collecting terms gives (12.2):

$$f^n(a, x) - x_0 = qS(x - x_0)^2 + pS(a - a_0) + O\big(S^2(|x - x_0|^4 + |a - a_0|^2)\big) + O\big(S(|x - x_0|^3 + |x - x_0||a - a_0|)\big).$$

As a consistency check, $f_x(x_0) = 0$ gives $\frac{1}{2}\,\partial^2 f^n/\partial x^2 = \frac{1}{2} S f_{xx}(x_0) = qS$, and $\partial f^n/\partial a = S f_a(x_0) + S p_1 = pS$. So $p = S^{-1}\,\partial f^n/\partial a\,(a_0, x_0)$, as the text states.

Step 6: The rescaled map converges to the canonical map

Use $y = qS(x - x_0)$ and $u = -pqS^2(a - a_0)$. The parameter is only rescaled, so the conjugate family is $g(u, y) = qS\big(f^n(a, x) - x_0\big)$, with

$$x - x_0 = \frac{y}{qS}, \qquad a - a_0 = -\frac{u}{pqS^2}.$$

Multiply (12.2) by qS term by term.

- *Leading terms.* $qS \cdot qS(x-x_0)^2 = \big(qS(x-x_0)\big)^2 = y^2$ and $qS \cdot pS(a-a_0) = pqS^2(a-a_0) = -u$. So the main part is $y^2 - u$.
- *First error.* $qS \cdot O\big(S^2(\xi^4+\alpha^2)\big) = O\big(S^3(\xi^4+\alpha^2)\big)$. With $\xi^4 \sim y^4/S^4$ and $\alpha^2 \sim u^2/S^4$ this is $O\big(S^{-1}(y^4+u^2)\big)$.
- *Second error.* $qS \cdot O\big(S(|\xi|^3+|\xi||\alpha|)\big) = O\big(S^2(|y|^3S^{-3}+|y||u|S^{-3})\big) = O\big(S^{-1}(|y|^3+|y||u|)\big)$.

On the region $|y|, |u| \le 2.5$ we have $y^4 \le 2.5|y|^3$, so everything combines to

$$g(u,y) = y^2 - u + O\big(S^{-1}(|y|^3+|y||u|+|u|^2)\big).$$

Validity. On this region $\xi = O(S^{-1})$ and $\alpha = O(S^{-2})$, while $x' - x_1 = O(S^{-2})$. All are small compared with S^{-1}, so the expansions of Step 4 apply.

Conclusion. By hypothesis (1), $S = O(r^n)$ with $r > 1$, so $S \to \infty$ and the error term tends to 0 uniformly on $[-2.5, 2.5]^2$. Hence g converges to the canonical map $y^2 - u$ as $n \to \infty$. All cascades of the canonical map lie in this bounded region, so the rescaled window of f^n reproduces the canonical window. By Steps 2 and 1, $R_f \to 9/4$. The sign of pq may reverse the direction of the parameter axis, and Step 2 shows this has no effect on the ratio. □

Numerical check from the text. In Table 12.3 for the Hénon map, the small windows of period 8 to 10 give $R_f \approx 2.245$ to 2.249, against $9/4 = 2.25$. The large windows fit worse, as expected, because hypotheses (1) to (3) are less well satisfied there.

13 Challenge 13: Box-Counting Dimension and Intersection

Setup. For $S \subset \mathbb{R}^m$ and $v \in \mathbb{R}^m$, $S + v = \{s + v : s \in S\}$. The translates of S_1 *almost never intersect* S_2 if $\{v : (S_1 + v) \cap S_2 \neq \varnothing\}$ has m-dimensional volume zero. Note that $(S_1 + v) \cap S_2 \neq \varnothing$ exactly when $v = s_2 - s_1$ for some $s_1 \in S_1$, $s_2 \in S_2$, so the bad set of translations is $S_2 - S_1$.

Steps 1 to 3 treat the middle-third Cantor set S, whose dimension $\ln 2/\ln 3 \approx 0.63$ satisfies $2 \cdot 0.63 > 1$, so Theorem 13.5 does not apply and intersections persist. Steps 4 to 6 treat the middle three-fifths set, where the dimensions sum to less than 1 and intersections are rare. Steps 7 to 9 prove Theorem 13.5.

Step 1: If $x \in S$ then $1 - x \in S$

S is the set of numbers in $[0, 1]$ with a base-3 expansion using only the digits 0 and 2. Since $1 = 0.222\ldots_3$, any $x = \sum_{i \ge 1} a_i 3^{-i}$ with $a_i \in \{0, 2\}$ satisfies

$$1 - x = \sum_{i \ge 1} (2 - a_i)\, 3^{-i}, \qquad 2 - a_i \in \{0, 2\}.$$

Subtracting from 1 swaps the digits $0 \leftrightarrow 2$, so $1 - x \in S$.

Step 2: Every $x \in [0,1]$ is a sum of two base-3 expansions using only the digits 0 and 1

Let $x = \sum x_i 3^{-i}$ with $x_i \in \{0,1,2\}$. Split each digit as $x_i = b_i + c_i$ with $b_i, c_i \in \{0,1\}$, using $0 = 0+0$, $1 = 1+0$ and $2 = 1+1$. Then

$$x = \sum b_i 3^{-i} + \sum c_i 3^{-i},$$

and both sums use only the digits 0 and 1. (For $x = 1 = 0.222\ldots_3$ the splitting is $0.111\ldots_3 + 0.111\ldots_3$.)

Step 3: $S + v$ meets S for every $v \in [0,1]$

For $v \in [0,1]$ the number $(v+1)/2$ lies in $[\frac{1}{2}, 1] \subset [0,1]$. By Step 2, $(v+1)/2 = B + C$, where B and C have base-3 digits only 0 and 1. Doubling a base-3 expansion digit by digit turns the digits $\{0,1\}$ into $\{0,2\}$ with no carries, so $s = 2B$ and $t = 2C$ belong to S, and

$$v + 1 = s + t.$$

By Step 1, $1 - t \in S$, and

$$v + (1 - t) = s \in S.$$

So the point $1 - t \in S$ satisfies $v + (1-t) \in S$, which says $(S + v) \cap S \neq \varnothing$. For $v \in [-1, 0]$ use the symmetry that $s + v = s'$ implies $s' + (-v) = s$, so $S + v$ meets S exactly when $S + (-v)$ does. Hence every translate with $-1 \leq v \leq 1$ meets S (and translates with $|v| > 1$ miss $S \subset [0,1]$ trivially).

Step 4: The middle-3/5 Cantor set $K(5)$ and its dimension

Digit description. At stage k the construction leaves 2^k closed intervals

$$I_{s_1 \cdots s_k} = \left[0.s_1 \cdots s_k,\ 0.s_1 \cdots s_k + 5^{-k}\right], \qquad s_i \in \{0, 4\}.$$

Removing the open middle three fifths of $I_{s_1 \cdots s_k}$ deletes the points whose next digit is 1, 2 or 3. It leaves the two intervals $I_{s_1 \cdots s_k 0}$ and $I_{s_1 \cdots s_k 4}$. The endpoints of the removed gap are $0.s_1 \cdots s_k 0444\ldots$ and $0.s_1 \cdots s_k 4000\ldots$, and they belong to the two surviving intervals. By induction, $K(5) = \bigcap_k E_k$ where E_k is the union of the stage-k intervals. A point lies in a nested sequence $I_{s_1} \supset I_{s_1 s_2} \supset \cdots$ exactly when it equals $\sum s_i 5^{-i}$, so $K(5)$ is the set of numbers in $[0,1]$ possessing a base-5 expansion with only the digits 0 and 4.

Dimension. Distinct stage-k intervals are separated by gaps of length at least $3 \cdot 5^{-k}$. A grid box of side 5^{-k} therefore meets at most one of them, and each interval is covered by at most two boxes. So

$$2^k \leq N(5^{-k}) \leq 2 \cdot 2^k, \qquad \frac{\ln N(5^{-k})}{\ln 5^k} \longrightarrow \frac{\ln 2}{\ln 5}.$$

For $\varepsilon \in [5^{-(k+1)}, 5^{-k}]$, monotonicity of $N(\varepsilon)$ sandwiches $N(\varepsilon)$ between $N(5^{-k})$ and $N(5^{-(k+1)})$, and the same limit results. Hence

$$d = \dim_{\text{box}} K(5) = \frac{\ln 2}{\ln 5} \approx 0.431.$$

If $S_1 = S_2 = K(5)$ then $d_1 + d_2 = \dfrac{2 \ln 2}{\ln 5} = \dfrac{\ln 4}{\ln 5} \approx 0.861 < 1$.

Step 5: If $v \in [0,1]$ has a base-5 expansion containing 22, then $S_1 + v$ misses S_2

Suppose, for contradiction, that $s_1 + v = s_2$ with $s_1, s_2 \in K(5)$. Write $s_1 = \sum a_i 5^{-i}$ and $s_2 = \sum b_i 5^{-i}$ with $a_i, b_i \in \{0, 4\}$. Then

$$v = \sum_{i \geq 1} d_i 5^{-i}, \qquad d_i = b_i - a_i \in D = \{-4, 0, 4\}.$$

For $j \geq 0$ define the tail $t_j = \sum_{i>j} d_i 5^{j-i}$. Then $|t_j| \leq 4 \sum_{k \geq 1} 5^{-k} = 1$ and $t_{j-1} = (d_j + t_j)/5$, so

$$5\, t_{j-1} \in D + [-1, 1] = [-5, -3] \cup [-1, 1] \cup [3, 5]. \tag{2}$$

In particular $5t_{j-1}$ never lies in the gaps $(1, 3)$ or $(-3, -1)$.

Now suppose some base-5 expansion $v = \sum v_i 5^{-i}$ has $v_j = v_{j+1} = 2$. Define the integers $M = \sum_{i<j} v_i 5^{j-1-i}$ and $N = \sum_{i<j} d_i 5^{j-1-i}$, and the fractional tail $u = \sum_{i \geq j} v_i 5^{j-1-i}$. Then

$$5^{j-1} v = M + u = N + t_{j-1}, \qquad \text{so} \qquad t_{j-1} - u = M - N \in \mathbb{Z}.$$

The digits $v_j v_{j+1} = 22$ force $u \in [\frac{12}{25}, \frac{13}{25}] = [0.48, 0.52]$, since the remaining digits contribute between 0 and $\frac{1}{25}$. As $t_{j-1} \in [-1, 1]$, the difference $t_{j-1} - u$ lies in $[-1.52, 0.52]$, so the integer $M - N$ is 0 or -1. Thus $t_{j-1} = u$ or $t_{j-1} = u - 1$, and in both cases

$$|5\, t_{j-1}| \in [2.4,\ 2.6],$$

which lies inside a gap excluded by (2). This contradiction shows $(S_1 + v) \cap S_2 = \varnothing$. The argument works for every expansion of v, including numbers that have two expansions.

Step 6: Translates of $K(5)$ almost never intersect $K(5)$

Let $\mathcal{N}$ be the set of $x \in [0, 1]$ having some base-5 expansion with no adjacent pair 22. Group the digits into consecutive blocks (v_{2k-1}, v_{2k}). If $x \in \mathcal{N}$ then none of its first n blocks equals 22. There are 25^n equally sized intervals of length 25^{-n} determined by the first n blocks, and 24^n of them have no block equal to 22. Every point of $\mathcal{N}$ lies in one of these 24^n closed intervals, so

$$\text{length}(\mathcal{N}) \leq 24^n \cdot 25^{-n} = \Big(\frac{24}{25}\Big)^n \longrightarrow 0.$$

(This is the hint's computation. The subinterval .22 occupies $\frac{1}{25}$ of $[0, 1]$, each remaining $.s_1 s_2$ contains a further $.s_1 s_2 22$ removed at the next stage, and the surviving lengths sum geometrically to 0.) So $\mathcal{N}$ has length zero.

By Step 5, if $v \in [0, 1]$ and $(K(5) + v) \cap K(5) \neq \varnothing$, then no expansion of v contains 22, so $v \in \mathcal{N}$. For $v \in [-1, 0)$ the symmetry of Step 3 reduces to $-v \in (0, 1]$, and for $|v| > 1$ there is no intersection because $K(5) \subset [0, 1]$. The set of bad translations is contained in $\mathcal{N} \cup (-\mathcal{N})$, which has length zero. Hence translates of the middle-3/5 Cantor set almost never intersect it. This agrees with Theorem 13.5, since $d_1 + d_2 < 1 = m$.

Step 7: Two ε-cubes rarely collide under translation

Let C_1, C_2 be m-dimensional cubes of side ε with centers P_1, P_2. Every point of such a cube lies within $\sqrt{m}\,\varepsilon/2$ of its center. If $(C_1 + v) \cap C_2 \neq \varnothing$, pick a common point z. Then

$$|P_1 + v - P_2| \leq |P_1 + v - z| + |z - P_2| \leq \sqrt{m}\,\varepsilon,$$

so v lies in the closed ball of radius $\sqrt{m}\,\varepsilon$ centered at $P_2 - P_1$. The cubes cannot meet while v stays outside that ball. The volume of the ball is $\omega_m m^{m/2}\varepsilon^m$, with ω_m the volume of the unit ball, so the set of colliding translations has volume at most $C\varepsilon^m$ with $C = \omega_m m^{m/2}$.

Step 8: The union over all pairs of cubes

Cover S_1 by $N_1(\varepsilon)$ grid cubes and S_2 by $N_2(\varepsilon)$ grid cubes. If $(S_1 + v) \cap S_2 \neq \varnothing$, there are $s_1 \in S_1$ and $s_2 \in S_2$ with $s_1 + v = s_2$. Let C_1 be the cube of the first cover containing s_1 and C_2 the cube of the second cover containing s_2. Then $(C_1 + v) \cap C_2 \neq \varnothing$, so v is a colliding translation for one of the $N_1(\varepsilon)N_2(\varepsilon)$ pairs. By Step 7 and subadditivity of volume, the set

$$B_\varepsilon = \{v \in B_m(r) : (C_1 + v) \cap C_2 \neq \varnothing \text{ for some pair}\}$$

has volume at most $C\,N_1(\varepsilon)N_2(\varepsilon)\,\varepsilon^m$, and the set of $v \in B_m(r)$ with $(S_1 + v) \cap S_2 \neq \varnothing$ is contained in B_ε.

Step 9: Passing to the limit

Fix $\delta > 0$ so small that $d_1 + d_2 + 2\delta < m$, which is possible because $d_1 + d_2 < m$. By the definition of box-counting dimension (with the limit superior if the limit does not exist), there is $\varepsilon_0 > 0$ such that for $\varepsilon < \varepsilon_0$,

$$N_i(\varepsilon) \le \varepsilon^{-(d_i+\delta)}, \qquad i = 1, 2.$$

Then

$$\operatorname{vol}(B_\varepsilon) \le C\,\varepsilon^{\,m-d_1-d_2-2\delta} \longrightarrow 0 \qquad (\varepsilon \to 0).$$

The set $\{v \in B_m(r) : (S_1 + v) \cap S_2 \neq \varnothing\}$ does not depend on ε and sits inside B_ε for every small ε, so it has outer measure zero and therefore volume zero. Since r was arbitrary, take $r = 1, 2, 3, \ldots$ and use countable subadditivity. The full set of translation vectors $v \in \mathbb{R}^m$ that produce an intersection has volume zero, so the translates of S_1 almost never intersect S_2. □

Postscript. The same counting idea underlies the embedding theorems. For distant pieces of an attractor of dimension d to miss each other in reconstruction space $\mathbb{R}^m$, the condition $2d < m$ of Theorem 13.5 is what is required.